Trade Policy Shocks and Market Reactions: An Event Study with LLM-Derived Measures of USTR Tariff Announcements

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BUS244

Context & Background

TSMC

- World's largest dedicated chip manufacturer
- Supplies leading technology companies with semiconductors

Technology Companies

- Main technology companies who rely on TSMC
- Apple, NVIDIA, AMD, Google, etc.

U.S. Trade Representative (USTR)

- Implementing tariff exclusions
- Section 301: gives USTR power to investigate unfair foreign trade policies

RQ: How do trade policy announcements affect semiconductor-related firm-level equity returns when exposure is indirect and supply-chain based?

Research Objectives & Contributions

Objectives

- ❑ Quantify how tariff announcements impact the performance of firms
- ❑ Measure abnormal returns
- ❑ Derive & compare LLM measures from USTR policy texts

Contributions

- ❑ Using an LLM to extract measures from policy text
- ❑ Differentiate types of policy implications using manual and LLM labels
- ❑ Implement generative AI in an event study

Outline

Data 01

Methodology 02

Results 03

Limitations & Further Work 04

Data: USTR Policy Text Corpus

USTR China Section 301

- ❑ Tariff Actions
- ❑ Exclusions
- ❑ Modifications

Format

- ❑ Event date
- ❑ Tariff list
- ❑ Document type

Transformations

- ❑ Clean text
- ❑ Ollama rationale text
- ❑ Action type
- ❑ Direction

trackage rights until January 31, 2020. NOPB Corp. filed the petition to partially revoke the exemption to allow the trackage rights to expire on that date.[2] NOPB Corp. argues that granting its petition will promote the rail transportation policy, will be consistent with the limited scope of the transaction, and will not adversely affect the current competitive situation of any shipper.

Discussion and Conclusions

Although NOPB Corp. and IC have expressly agreed on the duration of the proposed trackage rights agreement, trackage rights approved under the class exemption at 49 CFR 1180.2(d)(7) typically remain effective indefinitely, regardless of any contract provisions. Occasionally, however, the Board has partially revoked a trackage rights exemption to allow those rights to expire after a limited time period rather than lasting in perpetuity. *See, e.g., Ind. R.R.—Trackage Rights Exemption—CSX Transp., Inc.,* FD 36068 (Sub-No. 1) (STB served Feb. 9, 2017); *Ind. S. R.R.—Temporary Trackage Rights Exemption—Norfolk S. Ry.,* FD 35965 (Sub-No. 1) (STB served Nov. 25, 2015).

Under 49 U.S.C. 10502, the Board may exempt a person, class of persons, or a transaction or service, in whole or in part, when it finds that: (1) Continued regulation is not necessary to carry out the rail transportation policy of 49 U.S.C. 10101; and (2) either the transaction or service is of limited scope, or regulation is not necessary to protect shippers from the abuse of market power.

NOPB Corp.'s trackage rights were already authorized under the class exemption at 49 CFR 1180.2(d)(7) in Docket No. FD 36198. Granting partial revocation in these circumstances would promote the rail transportation policy by eliminating the need to file a second pleading seeking discontinuance when the agreement expires, thereby promoting rail transportation policy goals at 49 U.S.C. 10101(2), (7), and (15). Moreover, limiting the term of the trackage rights is consistent with the limited scope of the transaction previously exempted.[3] Therefore, the Board will grant the petition and permit the trackage rights exempted in Docket No. FD 36198 to expire on January 31, 2020.

To provide the statutorily mandated protection to any employee adversely affected by the discontinuance of trackage rights, the Board will impose the employee protective conditions set forth in *Oregon Short Line Railroad—Abandonment Portion Goshen Branch Between Firth & Ammon, in Bingham & Bonneville Counties, Idaho,* 360 I.C.C. 91 (1979).

This action is categorically excluded from environmental review under 49 CFR 1105.6(c).

It is ordered:

1. The petition for partial revocation is granted.
2. Under 49 U.S.C. 10502, the trackage rights described in Docket No. FD 36198 are exempted, as discussed above, to permit the trackage rights to expire on January 31, 2020, subject to the employee protective conditions set forth in *Oregon Short Line.*
3. Notice will be published in the **Federal Register** on June 20, 2018.
4. This decision is effective on July 20, 2018. Petitions to stay must be filed by July 2, 2018. Petitions for reconsideration must be filed by July 10, 2018.

Decided: June 13, 2018.

By the Board, Board Members Begeman and Miller.

Kenyatta Clay,

Clearance Clerk.

[FR Doc. 2018–13156 Filed 6–19–18; 8:45 am]

BILLING CODE 4915–01–P

[2] NOPB explains in its notice of exemption that because the extended trackage rights were to be more than a year in duration, the class exemption for temporary trackage rights under 49 CFR 1180.2(d)(8) was not available.

[3] Because the proposed transaction is of limited scope, the Board need not make a market power finding.

OFFICE OF THE UNITED STATES TRADE REPRESENTATIVE

[Docket Number USTR–2018–0018]

Notice of Action and Request for Public Comment Concerning Proposed Determination of Action Pursuant to Section 301: China's Acts, Policies, and Practices Related to Technology Transfer, Intellectual Property, and Innovation

AGENCY: Office of the United States Trade Representative.

ACTION: Notice of action, request for comments, and notice of public hearing.

SUMMARY: The U.S. Trade Representative (Trade Representative) has determined that appropriate action in this investigation includes the imposition of an additional ad valorem duty of 25 percent on products from China classified in the subheadings of the Harmonized Tariff Schedule of the United States (HTSUS) set out in Annex A of this notice. The Trade Representative has further determined to establish a process by which U.S. stakeholders may request that particular products classified within a covered tariff subheading in Annex A be excluded from these additional duties. Further, the Office of the U.S. Trade Representative (USTR) is seeking public comment and will hold a public hearing regarding a proposed additional action in this investigation. The proposed additional action is the imposition of an ad valorem duty of 25 percent on products of China classified in the HTSUS subheadings set out in Annex C of this notice.

DATES:

Applicable date of duties: The additional duties set out in Annex A to this notice are applicable with respect to products that are entered for consumption, or withdrawn from warehouse for consumption, on or after July 6, 2018.

Comment and hearing deadline: To be assured of consideration, you must submit comments and responses with respect to the proposed list of products in Annex C to this notice in accordance with the following schedule:

June 29, 2018: Due date for filing requests to appear and a summary of expected testimony at the public hearing and for filing pre-hearing submissions.

July 23, 2018: Due date for submission of written comments.

July 24, 2018: The Section 301 Committee will convene a public hearing in the main hearing room of the U.S. International Trade Commission, 500 E Street SW, Washington, DC 20436 beginning at 9:30 a.m.

July 31, 2018: Due date for submission of post-hearing rebuttal comments.

ADDRESSES: USTR strongly prefers electronic submissions made through the Federal eRulemaking Portal: *http://www.regulations.gov.* Follow the instructions for submitting comments in sections D, E, and F below. The docket number is USTR–2018–0018.

FOR FURTHER INFORMATION CONTACT: For questions about the ongoing investigation, action, or proposed additional action, contact USTR Assistant General Counsel Arthur Tsao at (202) 395–5725. For questions on customs classification or implementation of additional duties on products identified in Annex A to this Notice, contact *Traderemedy@cbp.dhs.gov.*

SUPPLEMENTARY INFORMATION:

A. Proceedings in the Investigation

On August 18, 2017, the Trade Representative initiated an investigation

Data: Ticker Universe

Firm level return data

- 01/2010 - present
  - TSMC
  - Semiconductors
  - Downstream tech
  - Energy/chemicals
  - Renewables
- Return estimations as percent changes

Methodology: Event Study

Final Results

Regression Analysis

LLM Classification

Market Model &
Abnormal Returns

Event Study:

- Financial analysis around events
- Understanding the effects before and after a particular event
  - Identifies causal market response to policy shocks
- Our case: event = tariff action

Methodology: Windows & Market Model

Estimation window

- Pre-event window used to estimate expected returns (-120 to -20 days)

Event window

- Window around event date used to measure abnormal price responses (-5 to +5 days)

Market model for expected returns

- Regression estimated separately for each event-firm pair
  - Over estimation window only
  - Coefficients used to generate expected returns during the event window

$$R_{i,t} = \alpha_i + \beta_i R_{m,t} + \epsilon_{i,t}$$

$R_{i,t}$ = return of firm i on day t

$R_{m,t}$ = market index return on day t

α_i, β_i = firm specific parameters estimated via OLS

$\epsilon_{i,t}$ = idiosyncratic return

Methodology: Abnormal Returns & CARs

Abnormal returns

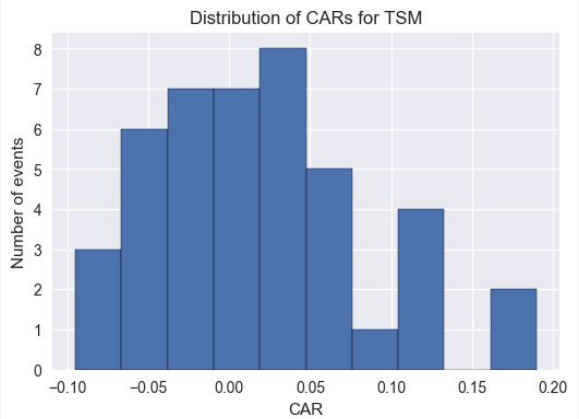
$$AR_{i,t} = R_{i,t} - \hat{R}_{i,t} \qquad \hat{R}_{i,t} = \hat{\alpha}_i + \hat{\beta}_i R_{m,t}$$

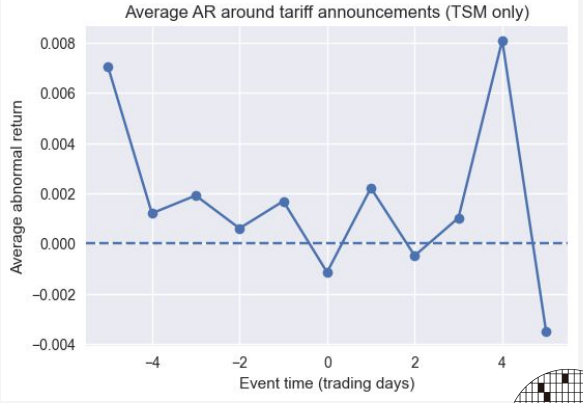
- ARs measure portion of firm's return that cannot be explained by general market movements
- Computed for each firm, day in event window
  - AR > 0 → outperformance, vice versa

Cumulative abnormal returns

$$CAR_i(\tau_1, \tau_2) = \sum_{t=\tau_1}^{\tau_2} AR_{i,t}$$

- Aggregation of ARs over the event window
- τ_1, τ_2 define the start and end of the event window





Methodology: LLM Classification

Ollama 3.2

- Four outputs
  - Severity score (0-3)
  - Surprise (0-1)
  - Direction
    - Tightening
    - Relief
    - Mixed
    - Unknown
  - Rationale

```
SYSTEM_PROMPT = """
You are a financial policy analyst. Given the full text of a USTR Section 301
tariff-related document, classify its expected impact on semiconductor-related firms.

You MUST return a JSON object with exactly these keys:
- "severity": integer in {0,1,2,3} where
    0 = no meaningful impact on semiconductors
    1 = mild impact
    2 = moderate impact
    3 = high / severe impact
- "surprise": float in [0,1] where higher = more surprising to markets
- "direction": one of {"tightening","relief","mixed","unknown"}
- "rationale": short natural-language explanation (2-3 sentences).

Return ONLY valid JSON. No extra commentary or text; just the JSON object.
"""
```

Methodology: LLM Classification Output

Summary of Events with LLM Columns

	event_id	file_name	list_id	doc_type	event_date	action_type	direction_guess	semiconductor_relevance	severity_label	direction	severity_ollama	surprise_ollama	direction_ollama	rationale_ollama
0	L1_noticeofaction_20180620	L1noticeofaction_20180620.pdf	1.000	noticeofaction	2018-06-20 00:00:00	imposition	tightening	True	3.000	tightening	2.000	0.400	relief	The proposed action is a partial revocation of trackage rights to expire after a limited time period, promoting rail transportation policy by eliminating the need for a second pleading. This decision is consistent with the limited scope of the transaction and will not adversely affect the current competitive situation of any shipper.
1	L2_noticeofaction_20180816	L2noticeofaction_20180816.pdf	2.000	noticeofaction	2018-08-16 00:00:00	imposition	tightening	True	3.000	tightening	2.000	0.600	tightening	The tariff increase on Chinese products will have a moderate impact on semiconductor-related firms due to the significant trade value of $16 billion and the broad scope of the tariffs, which affect various industries including technology and innovation.
2	L3_noticeofaction_20180921	L3noticeofaction_20180921.pdf	3.000	noticeofaction	2018-09-21 00:00:00	imposition	tightening	True	3.000	tightening	3.000	0.800	tightening	High impact due to significant tariff increase on products from China, affecting US exports and economy; the surprise is moderate as this action was expected by market participants.
3	L1_exclusions_20181228	L1exclusions_20181228.pdf	1.000	exclusions	2018-12-28 00:00:00	exclusions	relief	True	2.000	relief	2.000	0.500	relief	The U.S. Trade Representative established product exclusions for certain Chinese products, including semiconductor-related items, which will relieve some of the tariffs imposed on China.
4	L1_exclusions_20190325	L1exclusions_20190325.pdf	1.000	exclusions	2019-03-25 00:00:00	exclusions	relief	True	2.000	relief	2.000	0.700	relief	The USTR has granted product exclusions for certain Chinese products, including those used in semiconductors, reducing the scope of additional duties on China.

Full Summary of Events by Firm with All Measures (Top 5)

	event_id	ticker	event_trading_date	severity_label	direction	severity_ollama	surprise_ollama	direction_ollama	CAR_window	CAR	alpha_hat	beta_hat	n_est	n_ev
0	L1_noticeofaction_20180620	005930.KS	2018-06-20 00:00:00	3.000	tightening	2.000	0.400	relief	[-5,5]	-0.027	0.000	0.274	100.000	11.000
1	L1_noticeofaction_20180620	3576.TW	2018-06-20 00:00:00	3.000	tightening	2.000	0.400	relief	[-5,5]	-0.171	-0.001	-0.018	100.000	11.000
2	L1_noticeofaction_20180620	6505.TW	2018-06-20 00:00:00	3.000	tightening	2.000	0.400	relief	[-5,5]	-0.032	0.001	-0.119	100.000	11.000
3	L1_noticeofaction_20180620	AAPL	2018-06-20 00:00:00	3.000	tightening	2.000	0.400	relief	[-5,5]	-0.018	0.001	1.065	100.000	11.000
4	L1_noticeofaction_20180620	AMD	2018-06-20 00:00:00	3.000	tightening	2.000	0.400	relief	[-5,5]	-0.030	0.002	1.496	100.000	11.000

Methodology: Regression Analysis

Regressions

- Manual vs LLM
- Combined
- Severity index
  - Categorical labels → numerical index
  - Lower severity index = less severe tariff actions
- Sector fixed effects
- Cluster-robust standard errors
  - Ensures standard errors are clustered to adjust for correlations across firms reacting to the same announcement

$$CAR_{i,e,w} = \beta_0 + \beta_1 Severity_{e,m} + \beta_2 Direction_{e,m} + \epsilon_{i,e}$$

$$CAR_{i,e,w} = \beta_0 + \beta_1 Severity_{e,l} + \beta_2 Surprise_{e,l} + \beta_3 Direction_{e,l} + \epsilon_{i,e}$$

$$CAR_{i,e,w} = \beta_0 + \beta_1 Severity_{e,m} + \beta_2 Direction_{e,m} + \beta_3 Severity_{e,l} + \beta_4 Surprise + \beta_5 Direction_{e,l} + \epsilon_{i,e}$$

$$CAR_{TSMC,e,w} = \beta_0 + \beta_1 Severity\ Index_e + \beta_2 Direction_e + \epsilon_{i,e}$$

$$CAR_{all,e,w} = \beta_0 + \beta_1 Severity\ Index_e + \beta_2 Direction_e + \epsilon_{i,e}$$

$$CAR_{TSMC,e,w} = \beta_0 + \beta_1 Severity\ Index_e + \beta_2 Direction_e + Sector\ FE + \epsilon_{i,e}$$

Regression Results: Manual, LLM, Sector FE

OLS Regression Results

Dep. Variable:	CAR	**R-squared:**	0.006
Model:	OLS	**Adj. R-squared:**	-0.000
Method:	Least Squares	**F-statistic:**	31.06
Date:	Wed, 10 Dec 2025	**Prob (F-statistic):**	1.02e-08
Time:	12:49:23	**Log-Likelihood:**	354.68
No. Observations:	344	**AIC:**	-703.4
Df Residuals:	341	**BIC:**	-691.8
Df Model:	2		
Covariance Type:	cluster		

	coef	std err	z	P>|z|	[0.025	0.975]
const	-0.0025	0.002	-1.212	0.225	-0.007	0.002
severity_manual_num	-0.0010	0.003	-0.313	0.754	-0.007	0.005
dir_m_mixed	-0.0218	0.005	-4.666	0.000	-0.031	-0.013
dir_m_relief	0.0153	0.006	2.738	0.006	0.004	0.026
dir_m_tightening	0.0040	0.007	0.558	0.577	-0.010	0.018

Omnibus:	55.944	**Durbin-Watson:**	1.463
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	522.050
Skew:	-0.250	**Prob(JB):**	4.35e-114
Kurtosis:	9.014	**Cond. No.**	3.76e+15

Notes:
[1] Standard Errors are robust to cluster correlation (cluster)

Manual Labels

OLS Regression Results

Dep. Variable:	CAR	**R-squared:**	0.013
Model:	OLS	**Adj. R-squared:**	-0.008
Method:	Least Squares	**F-statistic:**	5.898
Date:	Wed, 10 Dec 2025	**Prob (F-statistic):**	0.000569
Time:	12:49:35	**Log-Likelihood:**	293.26
No. Observations:	288	**AIC:**	-572.5
Df Residuals:	281	**BIC:**	-546.9
Df Model:	6		
Covariance Type:	cluster		

	coef	std err	z	P>|z|	[0.025	0.975]
const	0.0153	0.008	1.951	0.051	-6.75e-05	0.031
severity_ollama_num	-0.0054	0.006	-0.923	0.356	-0.017	0.006
surprise_ollama_num	0.0102	0.021	0.487	0.626	-0.031	0.051
dir_llm_high	0.0302	0.007	4.392	0.000	0.017	0.044
dir_llm_mixed	-0.0191	0.012	-1.622	0.105	-0.042	0.004
dir_llm_relief	0.0061	0.005	1.160	0.246	-0.004	0.016
dir_llm_tightening	-0.0041	0.008	-0.489	0.625	-0.020	0.012
dir_llm_unknown	0.0021	0.008	0.271	0.787	-0.013	0.017

Omnibus:	57.806	**Durbin-Watson:**	1.379
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	522.392
Skew:	-0.462	**Prob(JB):**	3.66e-114
Kurtosis:	9.533	**Cond. No.**	9.37e+15

Notes:
[1] Standard Errors are robust to cluster correlation (cluster)

LLM Labels

OLS Regression Results

Dep. Variable:	CAR	**R-squared:**	0.021
Model:	OLS	**Adj. R-squared:**	0.003
Method:	Least Squares	**F-statistic:**	19.65
Date:	Wed, 10 Dec 2025	**Prob (F-statistic):**	2.78e-10
Time:	13:01:35	**Log-Likelihood:**	357.33
No. Observations:	344	**AIC:**	-700.7
Df Residuals:	337	**BIC:**	-673.8
Df Model:	6		
Covariance Type:	cluster		

	coef	std err	z	P>|z|	[0.025	0.975]
Intercept	-0.0293	0.007	-3.983	0.000	-0.044	-0.015
C(direction)[T.relief]	0.0371	0.004	8.437	0.000	0.028	0.046
C(direction)[T.tightening]	0.0271	0.011	2.532	0.011	0.006	0.048
C(sector)[T.Energy/Chemicals]	-0.0093	0.016	-0.573	0.567	-0.041	0.023
C(sector)[T.Other]	0.0151	0.009	1.661	0.097	-0.003	0.033
C(sector)[T.Renewables]	-0.0103	0.026	-0.392	0.695	-0.062	0.041
C(sector)[T.Semiconductors]	0.0153	0.009	1.741	0.082	-0.002	0.033
severity_num	-0.0022	0.005	-0.450	0.653	-0.012	0.008

Omnibus:	50.728	**Durbin-Watson:**	1.465
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	429.655
Skew:	-0.182	**Prob(JB):**	5.03e-94
Kurtosis:	8.463	**Cond. No.**	1.84e+15

Notes:
[1] Standard Errors are robust to cluster correlation (cluster)

Sector FE

Regression Results: Sector Fixed Effects

OLS Regression Results

Dep. Variable:	CAR	**R-squared:**	0.021
Model:	OLS	**Adj. R-squared:**	0.003
Method:	Least Squares	**F-statistic:**	19.65
Date:	Wed, 10 Dec 2025	**Prob (F-statistic):**	2.78e-10
Time:	13:01:35	**Log-Likelihood:**	357.33
No. Observations:	344	**AIC:**	-700.7
Df Residuals:	337	**BIC:**	-673.8
Df Model:	6		
Covariance Type:	cluster		

	coef	**std err**	**z**	**P>|z|**	**[0.025**	**0.975]**
Intercept	-0.0293	0.007	-3.983	0.000	-0.044	-0.015
C(direction)[T.relief]	0.0371	0.004	8.437	0.000	0.028	0.046
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C(sector)[T.Other]	0.0151	0.009	1.661	0.097	-0.003	0.033
C(sector)[T.Renewables]	-0.0103	0.026	-0.392	0.695	-0.062	0.041
C(sector)[T.Semiconductors]	0.0153	0.009	1.741	0.082	-0.002	0.033
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Skew:	-0.182	**Prob(JB):**	5.03e-94
Kurtosis:	8.463	**Cond. No.**	1.84e+15

Notes:
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Highlights:

- ❑ Direction: relief
- ❑ Direction: tightening
- ❑ Sector: other
- ❑ Sector: semiconductors

Results: Key Takeaways

- Relief announcements consistently generate positive ARs
- Mixed announcements produce negative & ambiguous market responses
  - Aligns with market uncertainty effects
- Severity/surprise scores were not statistically strong predictors of CARs
- Market reactions appear to be driven more by the uncertainty around events rather than the intensity of policy action
- Investors may interpret announcements in conjunction with information clarity/uncertainty

Discussions

Data Limitations

- Number of tariff events
  - Reduces statistical power
- Sensitivity of CARs to estimation windows
- Difference between event date and actual information arrival

LLM Limitations

- Ground truth in “severity” & “surprise” labels

Regression Limitations

- Linear effects only
- OVB & multicollinearity

Directions for Further Work

- Enhance text analysis pipeline
  - Embeddings vs categorical labels
- Domain specific fine-tuned models
- Expansion of policy text corpora & ticker universe
- Incorporating exogenous economic variables that better capture returns